

# TANMAY SHARMA

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## Education

### Northeastern University

Master of Science in Cybersecurity - 3.58/4.00

September 2024 – Present

Boston, MA

**Coursework:** Network Security, Linux Administration, Cryptography, Binary Exploitation, Shell Scripting, Security Risk Management and Assessment

### SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering - 3.56/4.00

September 2020 – July 2024

Chennai, India

**Coursework:** Operating Systems, Computer Architecture, Data Structures, Algorithms, Computer Networks

## Experience

### CyberspACe securiTY and forensIcs lab (CactiLab)

March 2024 – Present

Research Member

Boston, MA

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
- Analyzed FirmXRay software by OSUSecLab, focusing on baseaddresssolver.java module, to identify how base addresses are resolved and features extracted, improving vulnerability assessment capabilities in embedded firmware.

### Institute for Plasma Research (IPR)

September 2023 – November 2023

Network Systems Trainee - Cybersecurity

Gandhinagar, India

- Designed and deployed a secure, cost-effective network of Raspberry Pi-based access points, creating a dedicated wireless-to-wireless VPN/Wi-Fi solution for an isolated laboratory of 25–30 plasma physicists, enabling seamless and secure intra-lab communication.
- Implemented advanced encryption and security protocols, reducing network vulnerability by 60% and ensuring compliance with stringent research data protection standards, critical for sensitive plasma research experiments.
- Optimized network performance through real-time monitoring and custom configuration, achieving 99% uptime and improving data accessibility for researchers, which enhanced collaboration and accelerated experimental workflows by 30%.

## Projects

### Secure Instant Messaging System | *Python, SRP, ECDH, TCP Sockets*

January 2025 – April 2025

- Built a secure terminal-based chat system using SRP, ECDH, and AES-GCM for confidential, authenticated real-time communication (CY6740 - Network Security).
- Developed socket-based architecture with encrypted commands, rate limiting, and user listing, cutting abuse vectors by 80% and boosting resilience against attacks.

### DuckyScript Payloads for Flipper | *DuckyScript, HID Injection, Flipper Zero*

February 2025 – Present

- Engineered and maintained a comprehensive repository of advanced DuckyScript payloads and submodules for Flipper Zero, enabling HID-based attacks across Windows and Linux environments.
- Streamlined red team operations and ethical hacking workflows for the cybersecurity community by curating modular BadUSB exploits, credential harvesters, exfiltration scripts, and penetration testing utilities.

### Platform for Azure Response & Security Event Correlation | *Azure, AMA, Azure Sentinel*

November 2024

- Designed and deployed a cloud-based SIEM using Azure Sentinel, Log Analytics, and a Windows VM to collect and monitor security events in real time.
- Integrated Windows Security Events via Azure Monitor Agent (AMA), enabling centralized log analysis and scalable threat detection across virtual infrastructure.

## Certifications, Achievements & Community

**Certifications:** [CompTIA Security+](#) | [AWS Academy Cloud Security Foundations](#) | PNPT by TCM Security (In Progress)

**Achievements:** [SimSpace Cyber Cup 2025 - Rank 6](#), iCTF Rank 45 out of 1414

**Community:** Member of Boston Mesh Project and Boston Open Dev Community

## Technical Skills

**Languages:** C/C++, Python, Linux x86 Assembly (32bit/64bit), Bash, PowerShell, DuckyScript, SQL, YAML

**Security Tools:** Splunk, Azure Sentinel, Wireshark, Nmap, Sysmon, Ghidra, VirusTotal

**Technologies:** Cryptographic Protocols, SSL/TLS, IPsec, OAuth, Docker, Git, Linux Administration, Git

**Frameworks/Standards:** MITRE ATT&CK, NIST CSF, ISO 27001, OWASP Top 10, CMMC